

Acknowledgement

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Abstract

Social engineering remains one of the most effective attack vectors in modern cyber security, exploiting human psychology rather than technical vulnerabilities. With the advancement of artificial intelligence, attackers are now using sophisticated techniques such as deepfake audio/video, AI-generated phishing emails, and highly targeted spear-phishing campaigns. This report critically analyses modern social engineering methods, their evolution, forensic challenges, and detection strategies. A detailed forensic examination of phishing attacks and deepfake-based social engineering is presented, followed by an evaluation of their impact and the effectiveness of current countermeasures.

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1. Introduction

Social engineering has grown into one of the most dominant threats in cybersecurity, accounting for a significant percentage of breaches across organizations. Unlike traditional cyberattacks that rely on exploiting system vulnerabilities, social engineering manipulates human behaviour to deceive individuals into granting unauthorized access, leaking sensitive information, or performing harmful actions. Modern cybercriminals increasingly rely on advanced technologies such as artificial intelligence (AI), natural language generation, and deepfake tools to create highly convincing attacks (Dr. S. Gomathi Alias Rohini, 2025).

Phishing remains the most widespread form of social engineering, responsible for millions of dollars in financial loss each year. Meanwhile, deepfake technology powered by neural networks has enabled attackers to create realistic impersonations of executives, employees, or public figures. (Dr. S. Gomathi Alias Rohini, 2025).

This report critically explores the evolution of social engineering, focusing specifically on phishing and deepfake-based attacks. It provides technical insight into how such attacks are executed, the psychological principles behind them, and forensic methods used to detect, analyze, and prevent them (Dr. S. Gomathi Alias Rohini, 2025).

1.1 Aims and Objectives

The primary aim of this report is to critically examine modern social engineering attacks, with a particular focus on phishing and deepfake-based techniques, and to analyze the forensic methods used to detect, investigate, and mitigate such attacks in contemporary cyber security environments.

- To explore the concept of social engineering and its significance in modern cyber security threats.
- To critically analyze forensic techniques used in detecting and investigating phishing and deepfake-based attacks.
- To evaluate real-world case studies of social engineering incidents and assess their impact on organizations.

2. Background

Social engineering operates on the principle that humans are the weakest link in information security. Attackers exploit emotions such as fear, urgency, curiosity, or trust to manipulate users into performing desired actions. Traditionally, these attacks involved simple psychological tricks like fraudulent emails or misleading phone calls. However, the modern threat landscape shows an alarming increase in sophistication (Flechais, 2024).

See more at [appendix 8.3](#)

2.1 Case Study 1: Deepfake CEO Voice Fraud – The \$243,000 Financial Scam

2.1.1 Overview of Incident

In 2019, a UK-based energy company became a victim of one of the world's first publicly known deepfake audio social engineering attacks. The attackers used an AI-generated voice to impersonate the company's CEO and tricked the finance manager into transferring €220,000 (approx. \$243,000) to a fraudulent account.

The scammer mimicked the CEO's tone, accent, speech rhythm, and communication style so accurately that the employee believed the call was legitimate (Damiani, 2019).

See more at [appendix 8.1](#)



Figure 1: Deepfake voice call

2.2 Case Study 2: Social Engineering Leading to the 2023 MGM Cyberattack (Phishing + Vishing)

2.2.1 Overview of the incident

In September 2023, MGM Resorts International suffered one of the most disruptive cyberattacks in hospitality industry history. Attackers from the hacking group ALPHV/BlackCat infiltrated MGM's internal systems by using simple social engineering techniques—not malware, not zero-days.

They impersonated an MGM employee over the phone and convinced IT helpdesk staff to reset credentials, leading to system-wide outages for 10+ days and millions of dollars in losses (Newman, 2023).

See more at [appendix 8.2](#)



Figure 2: MGM casino hack

3. Method, Opportunity and Motive

3.1 Method

Attackers typically follow a structured method:

- **Reconnaissance:** collect victim information via OSINT, social media
- **Crafting malicious content:** phishing email, fake website, or deepfake audio
- **Delivery:** email, call, messaging apps, video conferencing
- **Execution:** victim clicks link, shares credentials, or transfers money
- **Covering Tracks:** deleting logs, using VPNs, proxies, burner accounts

Deepfake attacks add an additional step of AI model training using victim voice or images (Alex Mathew, 2025).

3.2 Opportunity

Opportunities arise due to:

- weak cybersecurity awareness
- lack of multi-factor authentication (MFA)
- remote work culture
- widespread use of social media
- availability of AI/ML tools to create fake content

3.3 Motive

Motives commonly include:

- financial gain (BEC scams, fraud)
- espionage / intelligence collection
- unauthorized access to systems
- reputational damage
- identity theft

For example, deepfake voice phishing has been used to trick organizations into transferring large sums of money to attacker-controlled accounts (Alex Mathew, 2025).

4. Critical Forensic Analysis

This section explores how forensic investigators detect and analyse deepfake-based social engineering attacks.

4.1 Environment Setup

4.1.1 Setting up Environment

To conduct the critical analysis of a deepfake-assisted phishing scenario, a controlled and isolated test environment was established. A virtualized laboratory setup was created using VirtualBox, with Ubuntu selected as the guest operating system.

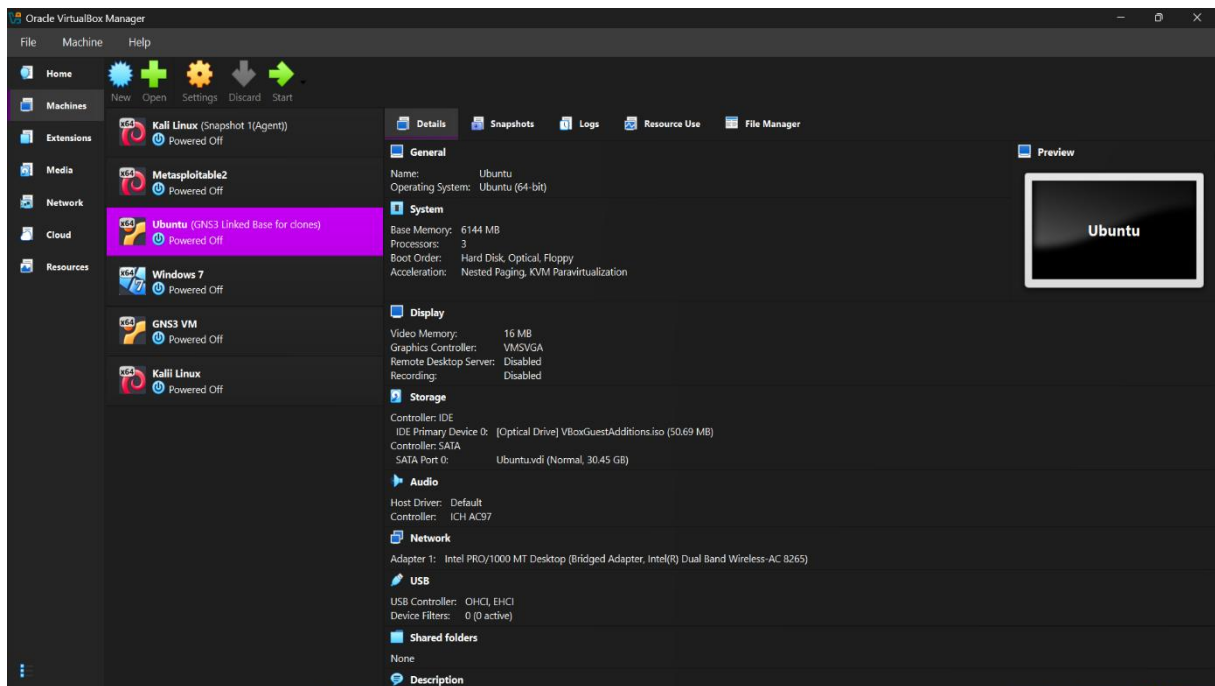


Figure 3: VM with ubuntu setup

4.1.2 Installing Zphisher Tool

Zphisher is an open-source social-engineering simulation tool hosted on GitHub and is commonly referenced in cybersecurity education to demonstrate how credential-harvesting attacks operate.

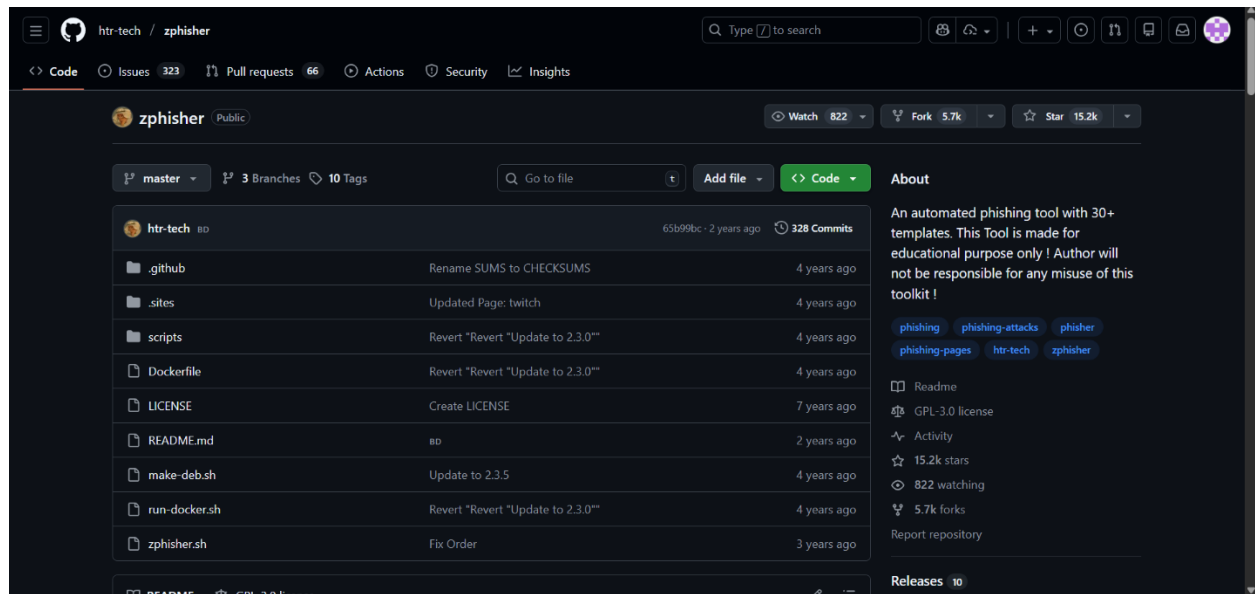


Figure 4: Zphisher on Github

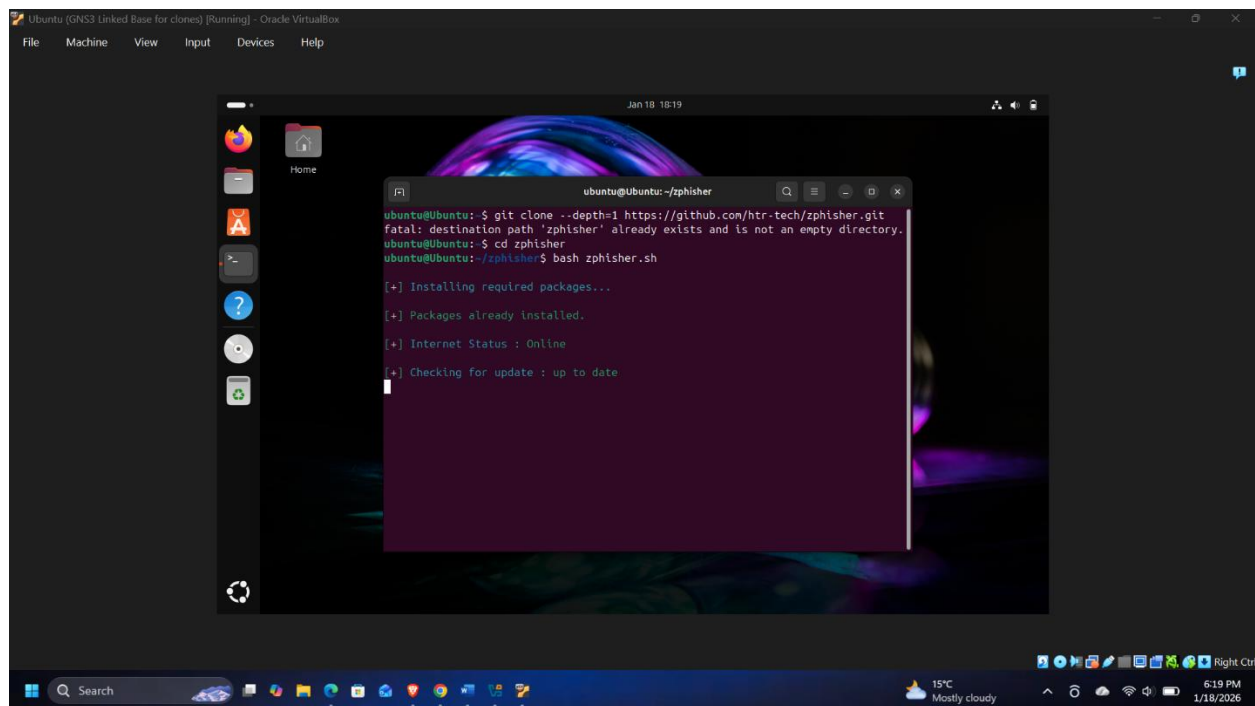


Figure 5: Installing zphisher on ubuntu

4.1.3 AI-Generated Deepfake video

The video was produced using HeyGen, a web-based artificial intelligence platform that allows users to generate realistic synthetic videos using pre-trained digital avatars and voice models. The objective of creating the deepfake video was not to deceive real users, but to critically analyse how AI-generated media can be abused within phishing and social-engineering attacks. The video featured a synthetic representation of Elon Musk, selected due to his high public visibility and perceived authority in the technology and financial sectors. Attackers frequently exploit such public figures to increase credibility, making this choice relevant for analysis.

The AI-generated video presented a fabricated announcement claiming to be related to a security upgrade for PayPal users. The narrative followed a common phishing pattern observed in real-world incidents, where victims are informed of urgent security changes requiring immediate action.

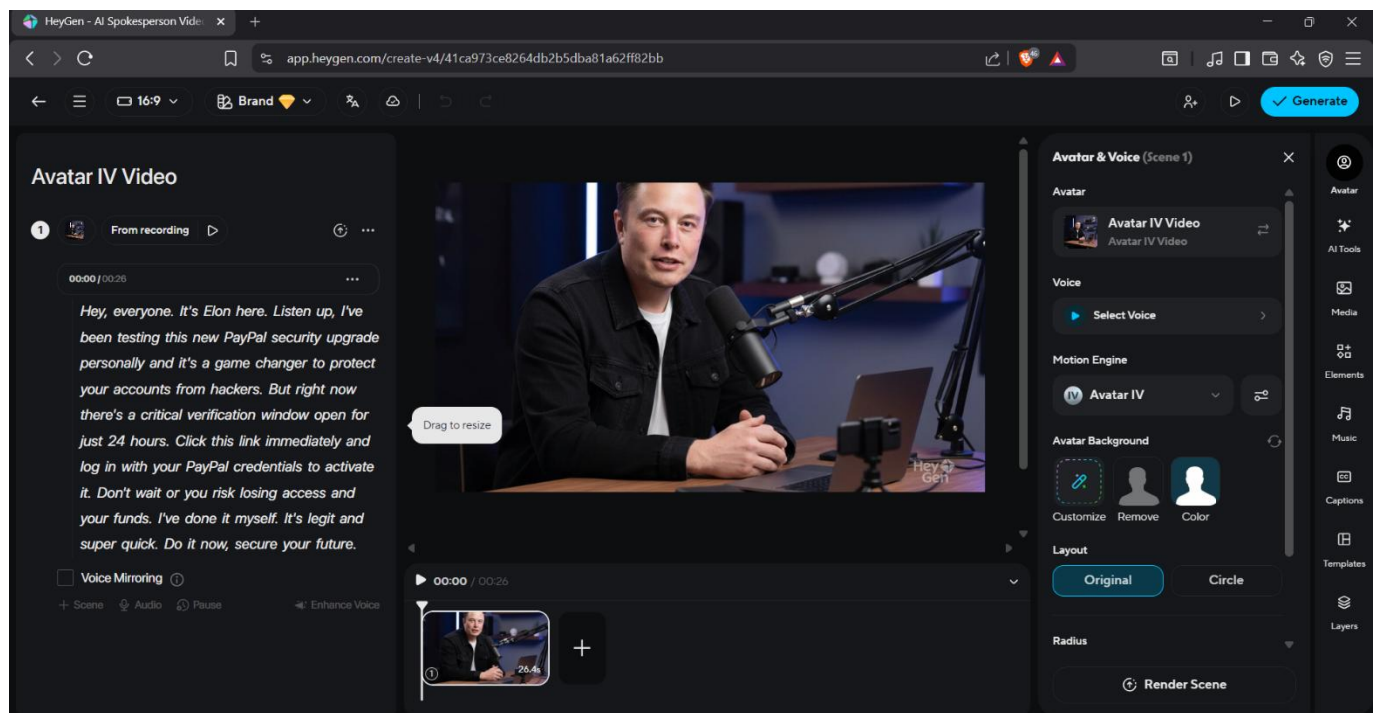


Figure 6: Creating AI video

To watch full video, click this link:

<https://drive.google.com/file/d/1KA2oo8a2zsTISlrWcalHsfid9L3nYQ5j/view?usp=sharing>

4.2 Attack Demonstration

4.2.1 Overview of Attack Scenario

This attack simulation demonstrates a social-media-based phishing scenario enhanced by AI-generated deepfake content. Unlike traditional phishing campaigns that rely on mass email distribution, this simulation focuses on visual trust exploitation through social media, reflecting a growing real-world threat trend.

The simulated attack involved posting an AI-generated deepfake video on Instagram as a story, accompanied by a link directing the viewer to a fake login page. The goal was to evaluate how synthetic authority, platform trust, and urgency influence user behaviour.

4.2.2 Link Generating using Zphisher

The generated link pointed to a cloned authentication page created using Zphisher, which served as a controlled mechanism for observing how users respond when prompted to verify account details.

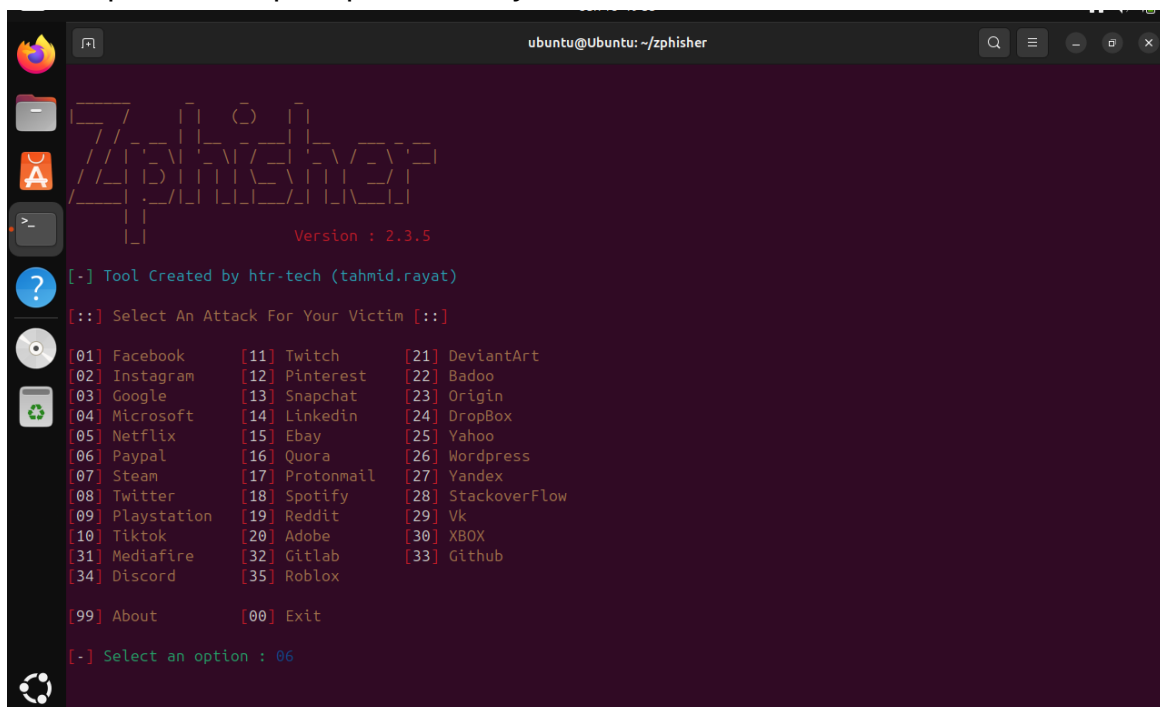


Figure 7: Choosing Paypal option

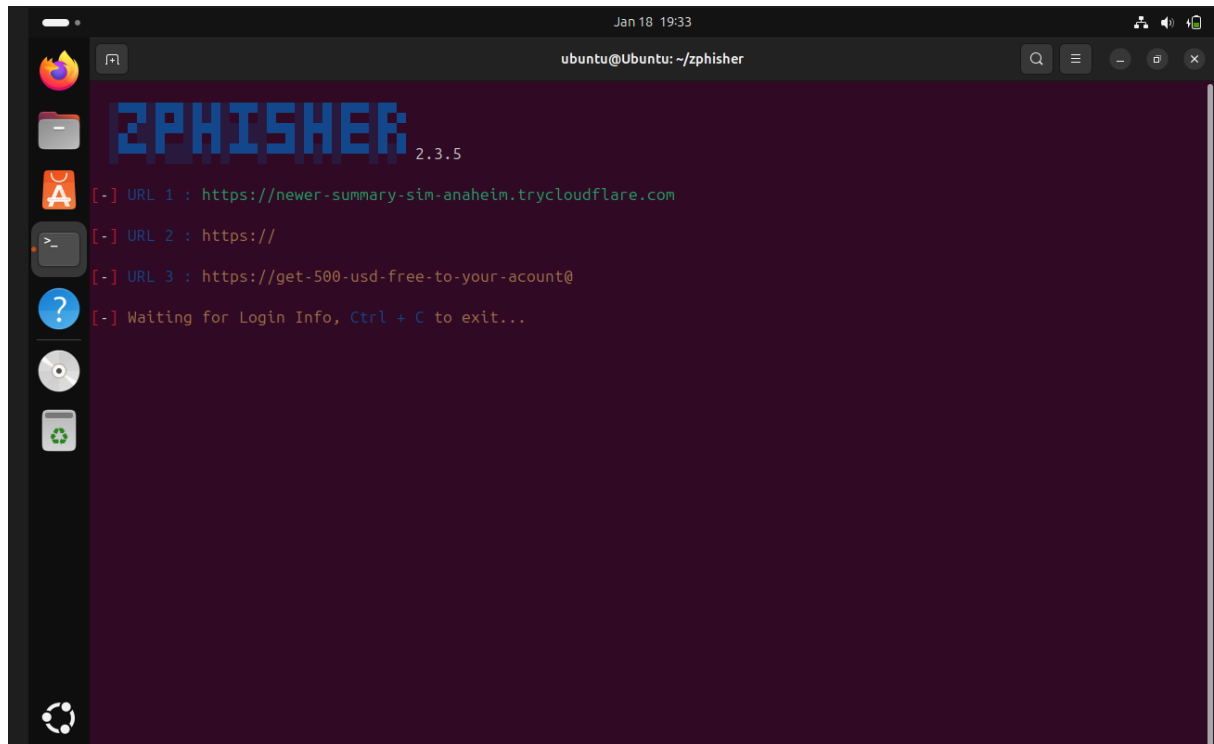
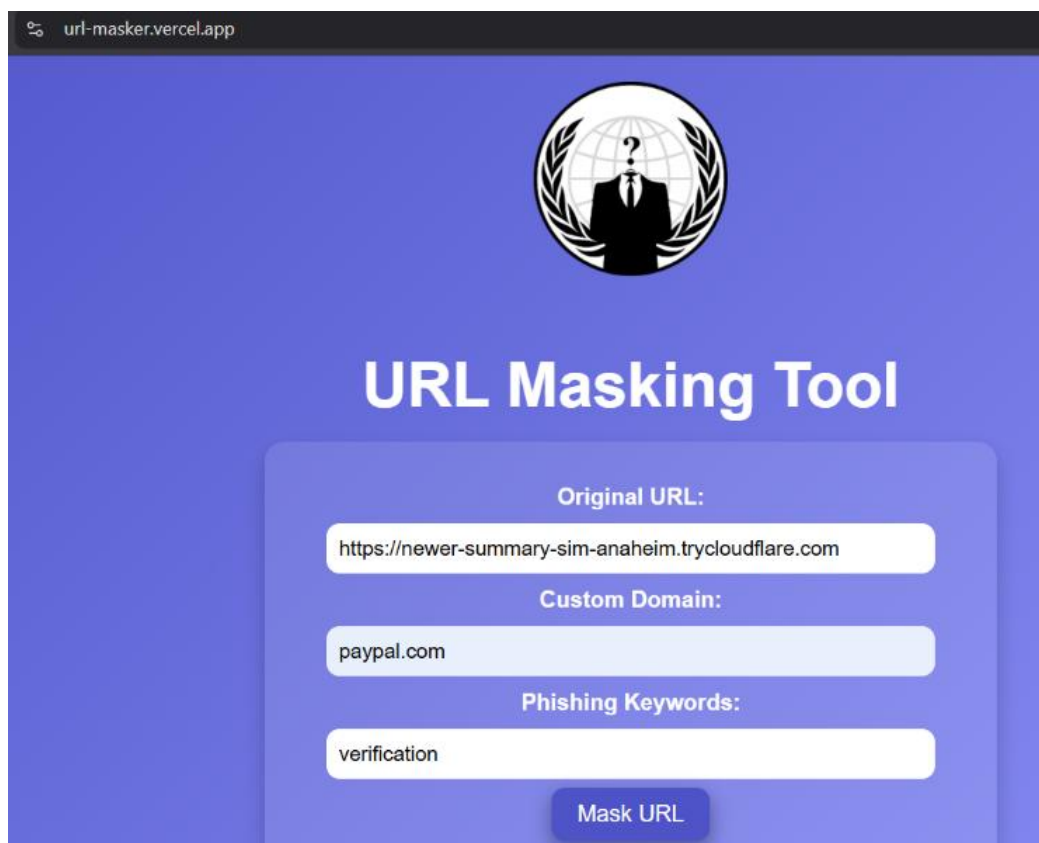


Figure 8: Link generated

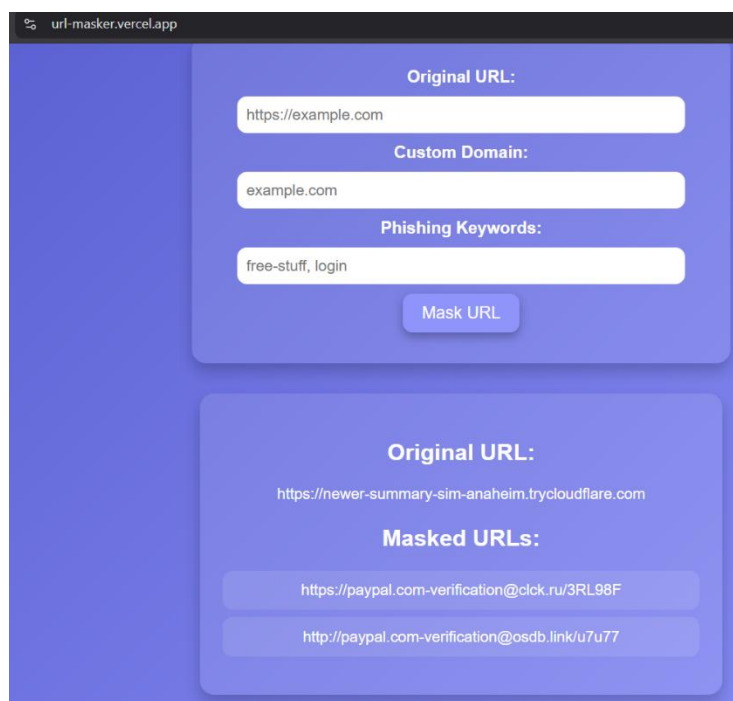
4.2.3 URL Masking

URL masking does not change the destination of a link but modifies how it appears to the user. From an analytical perspective, this technique demonstrates how attackers attempt to reduce suspicion by hiding unfamiliar or technical-looking URLs behind cleaner or more reassuring text. I used URL Masker website for masking URL.



The screenshot shows the URL Masking Tool interface. At the top, there is a logo featuring a globe with a question mark and a silhouette of a person in a suit. Below the logo, the title "URL Masking Tool" is displayed in large white text. The interface contains three input fields: "Original URL:" with the value "https://newer-summary-sim-anaheim.trycloudflare.com", "Custom Domain:" with the value "paypal.com", and "Phishing Keywords:" with the value "verification". A blue button labeled "Mask URL" is positioned below the input fields.

Figure 9: Entered link generated from zphisher



The screenshot shows the URL Masking Tool interface after processing. The input fields are the same as in Figure 9. Below the input fields, a section titled "Masked URLs:" displays two generated URLs: "https://paypal.com-verification@clk.ru/3RL98F" and "http://paypal.com-verification@osdb.link/u7u77".

Figure 10: Get the masked url

4.2.4 Delivery via Instagram Story

The deepfake video was posted as an Instagram story with the link attached. Social media stories are effective for attackers because users are more likely to click without careful checking and also expect quick content.

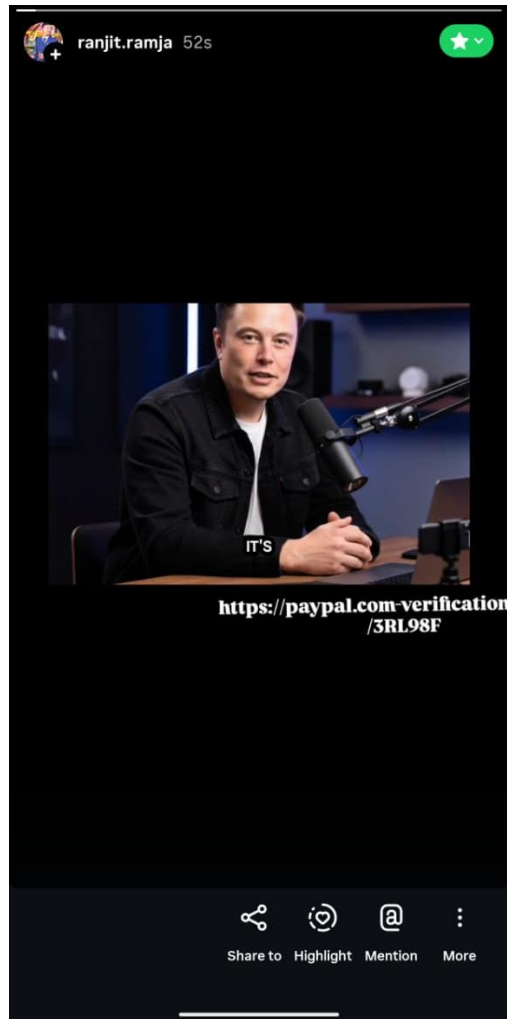


Figure 11: Posted on insta story

4.2.5 Redirection to a Fake Authentication Interface

The link shared within the story redirected the user to a simulated login page generated using Zphisher. The page visually resembled a legitimate service interface to maintain consistency with the video narrative.

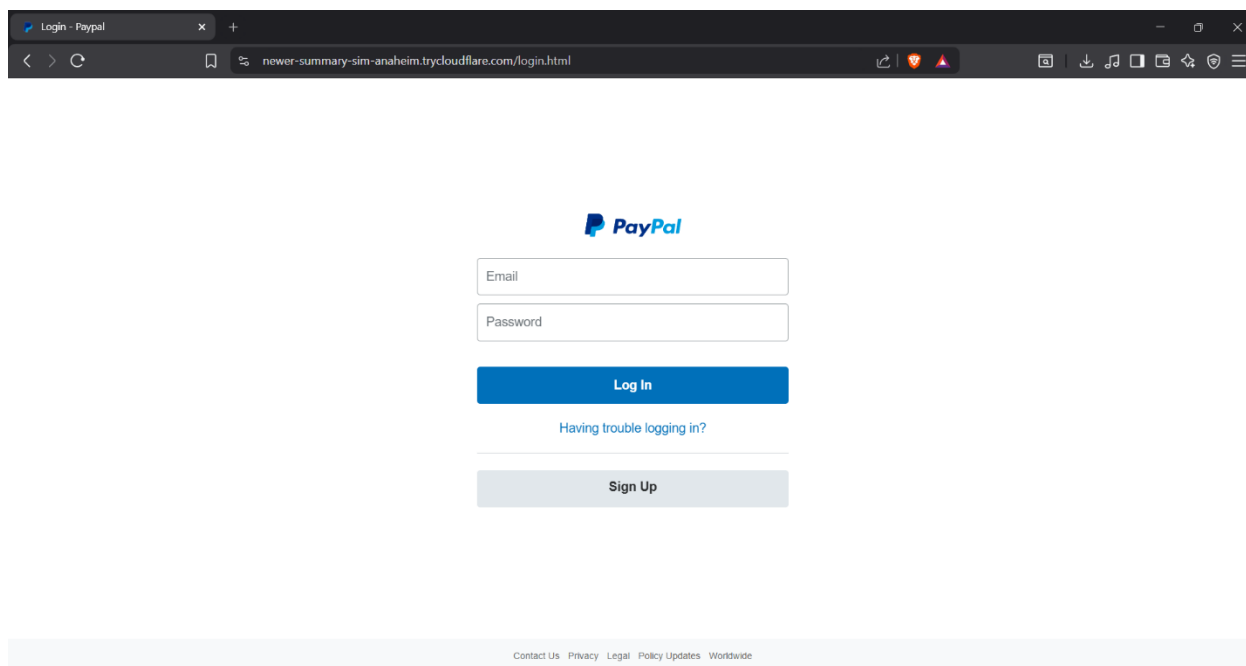


Figure 12: Clone of Paypal

4.2.6 Credential Submission

When test credentials were entered into the fake login interface, they were recorded for analytical observation only.

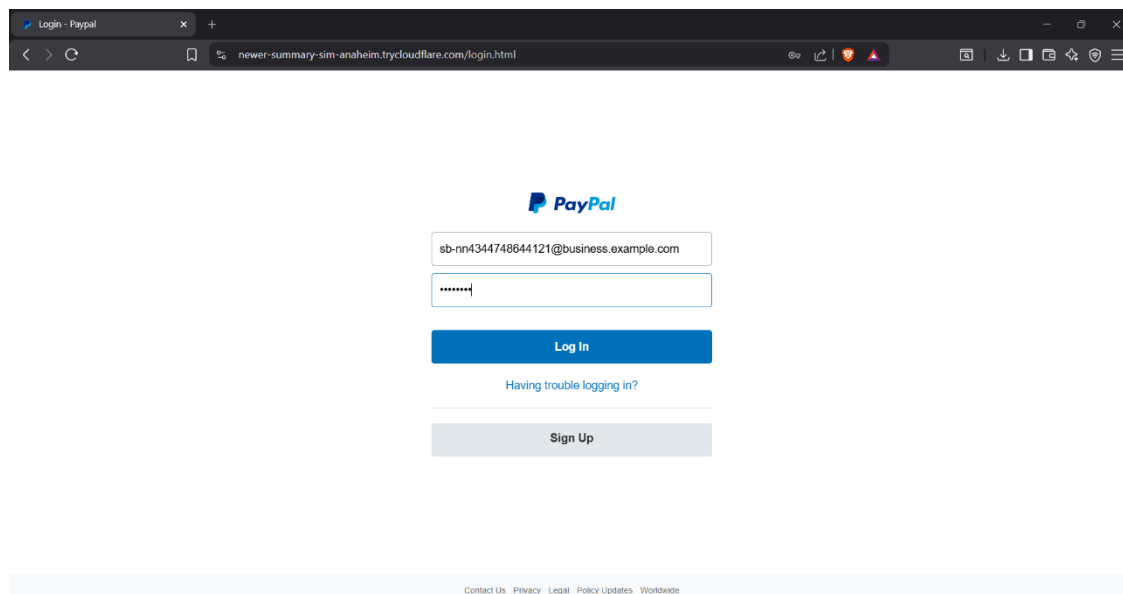
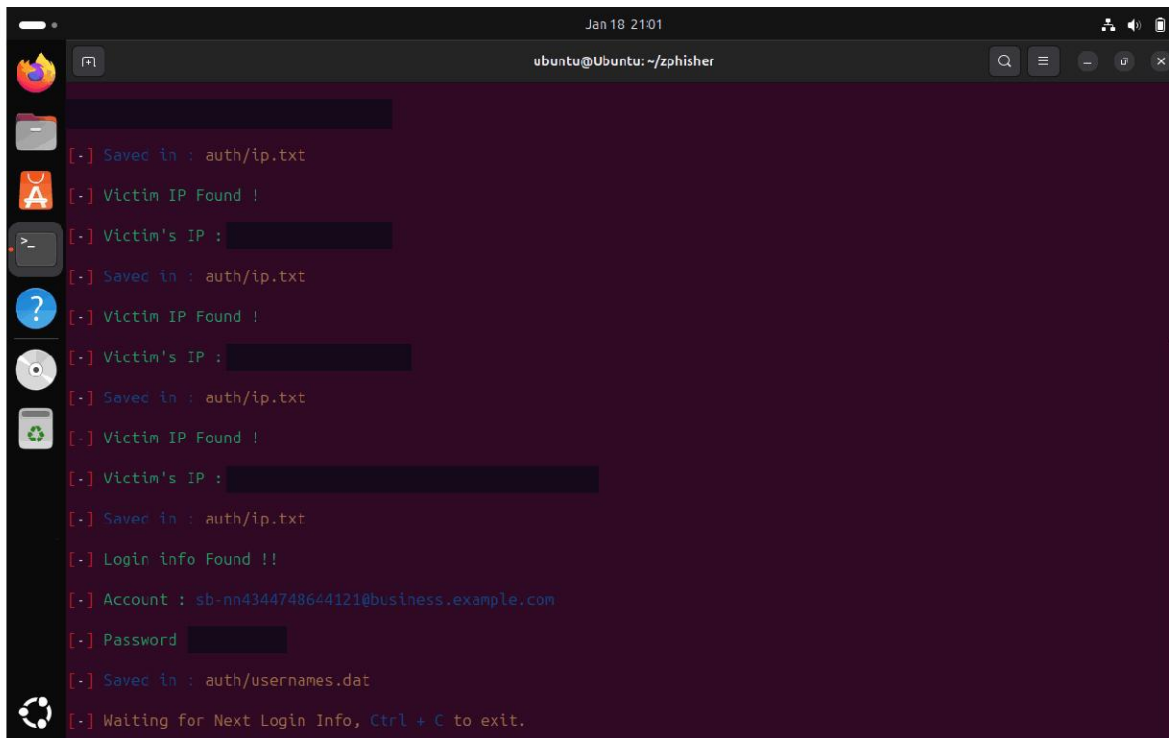


Figure 13: Entered credential



```
Jan 18 21:01
ubuntu@Ubuntu: ~/zphisher

[+] Saved in : auth/ip.txt
[+] Victim IP Found !
[+] Victim's IP : 
[+] Saved in : auth/ip.txt
[+] Victim IP Found !
[+] Victim's IP : 
[+] Saved in : auth/ip.txt
[+] Victim IP Found !
[+] Victim's IP : 
[+] Saved in : auth/ip.txt
[+] Login info Found !!
[+] Account : sb-nn4344748644121@business.example.com
[+] Password 
[+] Saved in : auth/usernames.dat
[+] Waiting for Next Login Info, Ctrl + C to exit.
```

Figure 15: Successfully captured credential

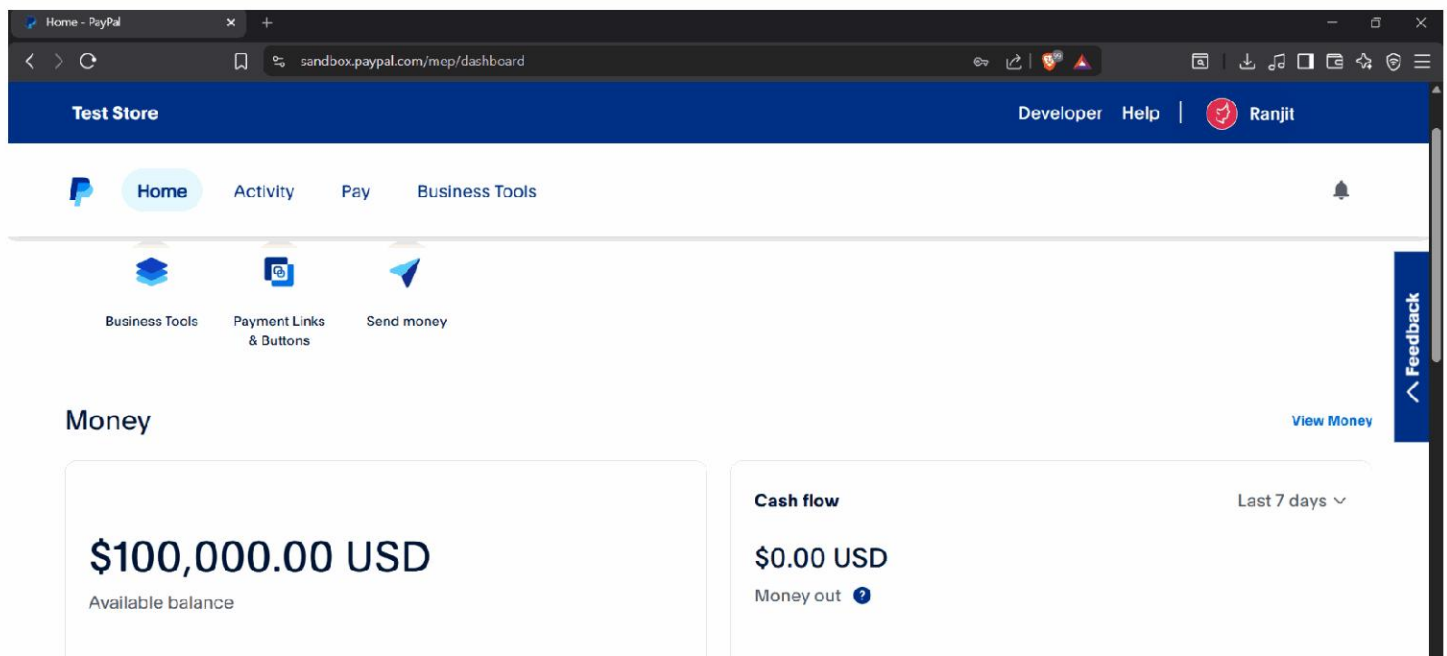


Figure 14: Successfully logged in

4.3 Analysis of Simulated Attack

This forensic analysis examines a social-media phishing simulation enhanced by an AI-generated deepfake video. The objective is to reconstruct what happened, identify what evidence is produced at each stage, and extract indicators that would help a defender detect or investigate a similar incident.

See more at [appendix 8.4](#)

5. Results and Evaluation

This section evaluates the outcomes of the simulated social engineering attack and the forensic analysis conducted throughout the investigation. The aim is to assess the effectiveness of modern phishing and deepfake-based techniques, as well as the reliability of forensic methods used to detect and analyse such attacks.

5.1 Results of the Simulated Attack

The results of the simulated phishing attack demonstrated that the combination of AI-generated deepfake media and social engineering techniques can significantly increase the credibility and success rate of phishing campaigns. The Instagram-based delivery method proved particularly effective, as users tend to consume social media content quickly and with limited analysis.

The attack simulation successfully achieved the following outcomes:

- The AI-generated deepfake video gained immediate attention due to the impersonation of a well-known public figure
- The use of urgency and authority encouraged users to interact with the embedded link without verification
- The masked URL reduced suspicion by concealing the actual destination domain
- The cloned authentication interface successfully captured test credentials, demonstrating how easily sensitive information can be harvested when visual trust is exploited

These results highlight how modern phishing attacks no longer rely solely on poorly written emails but instead use multimedia, platform trust, and psychological manipulation to deceive victims.



Figure 16: Credential Harvesting

5.2 Evaluation

The evaluation of attack shows that human factors remain the weakest link in cybersecurity. Despite advancements in firewalls, intrusion detection systems, and authentication controls, social engineering attacks continue to succeed by exploiting trust and urgency.

The forensic analysis applied during this investigation proved effective in identifying multiple indicators of compromise. AI detection tools such as Zhuque AI Detection Assistant and Sightengine successfully flagged the video as synthetic, based on facial inconsistencies, motion artefacts, and unnatural visual patterns. This demonstrates that automated deepfake detection tools can play a critical role in early identification of AI-generated media.

URL analysis using urlscan.io revealed that the masked link redirected to an untrusted domain unrelated to the legitimate PayPal infrastructure. Comparing the fake URL with the official PayPal domain clearly demonstrated domain impersonation, a key indicator of phishing activity. From a digital forensics perspective, the investigation confirmed that:

- Social media content can act as primary digital evidence in modern phishing cases
- Metadata, URL structure, and hosting information remain crucial forensic artefacts
- AI-generated content introduces new challenges, as visual realism can bypass human judgment.

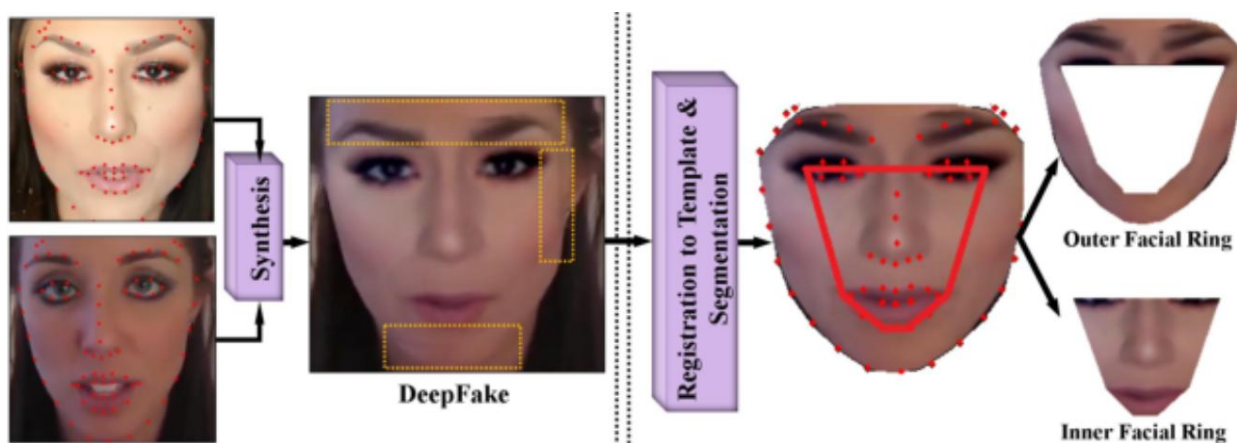


Figure 17: Deepfake Detection

6. Conclusion

This report critically examined social engineering in modern cyber security, with a specific focus on phishing and deepfake-based attacks. Through detailed background research, real-world case studies, and a controlled attack simulation, the study demonstrated how attackers exploit human psychology, trust in authority, and emerging AI technologies to bypass traditional security controls.

The analysis of the deepfake CEO voice fraud and the MGM cyberattack illustrated that even well-resourced organisations can be compromised through simple yet highly effective social engineering techniques. These cases highlight that technical security measures alone are insufficient when human factors are not adequately addressed.

The simulated attack further demonstrated how AI-generated media, combined with social media platforms and credential-harvesting tools, can significantly increase attack success rates. The forensic investigation showed that although detection tools can identify synthetic media and phishing indicators, such attacks remain difficult to detect at an early stage without specialised knowledge and tools.

In conclusion, social engineering remains one of the most dangerous and evolving threats in cyber security. Organisations must adopt a holistic defence strategy that combines technical controls, forensic preparedness, employee awareness training, and zero-trust identity verification. As AI-generated attacks continue to advance, digital forensics will play an increasingly critical role in detecting, analysing, and mitigating future cybercrime incidents.

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8. Appendix

8.1 Case Study 1

8.1.1 Attack Breakdown

Step 1: Reconnaissance (OSINT)

Attackers gathered detailed intelligence:

- CEO's voice from interviews, webinars, and earnings calls
- Company structure
- Finance manager's role and responsibility
- Usual communication patterns

Step 2: Deepfake Voice Synthesis

Using a Generative Adversarial Network (GAN) or voice cloning model (e.g., Lyrebird, Descript Overdub), attackers created a high-fidelity voice clone.

Step 3: Social Engineering Execution

The finance manager received a call that:

- Sounded identical to the CEO
- Contained urgency
- Requested a "confidential, time-sensitive" bank transfer
- Was reinforced by a spoofed email to appear legitimate

Step 4: Fraud Completion

The victim transferred the funds, only realizing the scam after a second fraudulent request raised suspicion (Sloane, 2019).

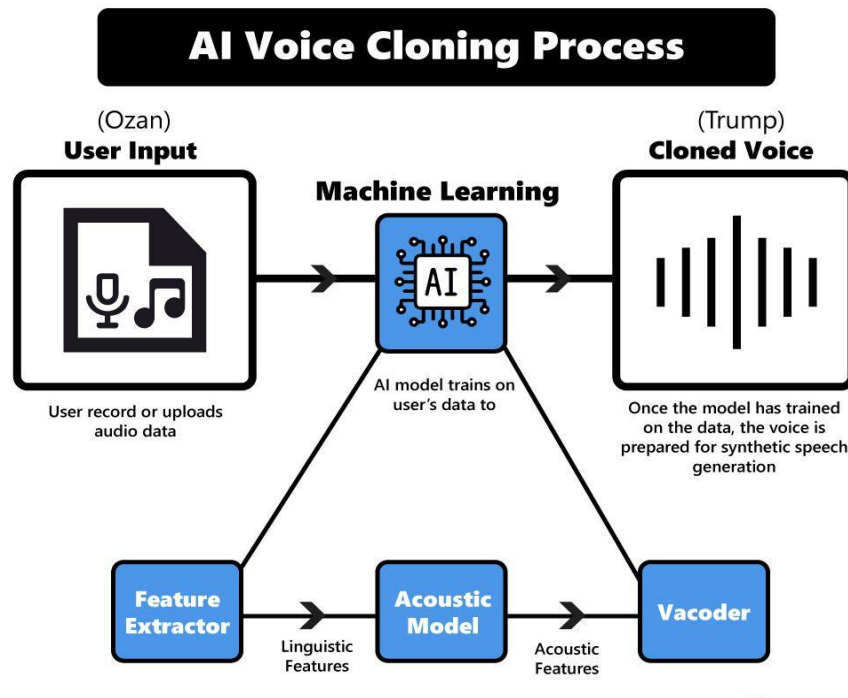


Figure 18: AI voice cloning process

8.1.2 Forensic Analysis

1. Voiceprint Examination

Forensic analysts observed:

- Unnatural harmonic patterns in the spectrogram
- Repetitive noise artifacts
- Inconsistent emotional modulation
- Micro-pauses typical of AI-generated speech

2. Network Forensics

- VoIP call routed via multiple countries
- IP addresses linked to anonymization services
- No physical call metadata (SIM card, telecom provider)

3. Email Header Analysis

- SPF/DKIM mismatches
- Slight domain spoofing
- Routing anomalies confirming phishing origin

4. Financial Forensics

- Money transferred to a Hungarian account
- Immediately moved to multiple offshore accounts
- Classic money laundering via “smurfing” technique (Cinder, 2019)

8.1.3 Self-Analysis

1. Why this attack succeeded:

- High trust relationship between CEO and finance manager
- Lack of verification protocol for unusual transactions
- Deepfake audio realism bypassed human judgment
- Urgency exploited emotional decision-making

2. What I learned:

- This case shows that deepfake social engineering has evolved beyond simple scams into highly targeted, multi-layered attacks. Even trained employees can be tricked when psychological pressure and AI-generated media work together.

3. What organizations must do:

- Implement multi-person authorization (4-eyes principle)
- Use code phrases for high-value approvals
- Introduce deepfake awareness training
- Monitor VoIP anomalies using behavioural analytics (ICAEW Insights, 2025).

Back at [Case Study 1](#)

8.2 Case Study 2

8.2.1 Attack Breakdown

Step 1: Information Gathering

Attackers used LinkedIn to collect:

- Employee names
- Job roles
- Department hierarchy
- Working hours

Step 2: Impersonation & Vishing

- They called the MGM helpdesk pretending to be an employee needing a password reset. Key tactics used:
- Confident and authoritative tone
- Technical language used casually
- Social pressure (claiming urgent work issue)

Step 3: Privilege Escalation

- With the new credentials, attackers:
- Accessed MGM's Okta environment
- Deployed ransomware
- Disabled systems
- Encrypted critical servers

Step 4: Business Impact

- Slot machines shut down
- Digital hotel room keys failed
- Reservation systems crashed
- Guests locked out of rooms
- MGM lost >\$100 million in revenue (RITTER, 2023).



Figure 19: Notice from MGM resort after hack

8.2.2 Forensic Analysis

1. Social Engineering Trail

- Call recordings reviewed
- Voice metadata indicated VoIP spoofing
- No malware used during initial entry
- Attack fully depended on the helpdesk's trust

2. Identity Verification Failures

Forensic assessment found:

- Helpdesk lacked strict identity confirmation
- No MFA challenge before password reset
- Overreliance on verbal confirmation

3. Log Analysis

System logs revealed:

- Multiple unusual login attempts
- Lateral movement across internal servers
- Rapid privilege escalation
- Use of legitimate admin tools (Living-off-the-Land)

4. Ransomware Deployment Patterns

Investigators found:

- ALPHV/BlackCat ransomware encrypted VMs
- Attackers used PowerShell to disable endpoint protection
- RDP sessions opened shortly after credential compromise (Schrader, 2025)

8.2.3 Self-Analysis

A. Why this attack was successful:

- MGM relied heavily on human-based authentication
- Helpdesk employees were not trained for modern phishing tactics
- Social engineering bypassed all cyber defenses
- Attackers needed only a 10-minute phone call to take down an entire corporation

B. What I learned:

This case proves that even the biggest companies with advanced firewalls, SIEMs, and endpoint protection can fall victim to simple social engineering. It highlights the idea that cybersecurity is only as strong as the least prepared employee.

C. What organizations must do:

- Strict multi-factor verification for password resets
- Zero-trust identity management
- Regular phishing simulation exercises
- Automated helpdesk workflows that reduce human judgment errors (Schrader, 2025)

Back at [Case Study 2](#)

8.3 Background

8.3.1 Evolution of Phishing Attacks

Phishing has evolved from generic mass emails to advanced techniques such as:

- **Spear Phishing:** personalized attacks aimed at specific individuals
- **Whaling:** targeting high-profile executives (CEO, CFO, etc.)
- **Clone Phishing:** replicating legitimate emails
- **Business Email Compromise (BEC):** impersonating business partners or executives
- **Smishing/Vishing:** phishing through SMS or voice calls

Phishing kits available on the dark web allow even low-skilled attackers to launch highly convincing campaigns (Flechais, 2024).



Figure 20: Phishing Attack

8.3.2 Rise of Deepfake-Based Social Engineering

Deepfakes use AI models (GANs, autoencoders) to synthesize realistic human speech, faces, and videos. Attackers now use these tools to:

- impersonate CEOs in video calls
- generate voice calls authorizing fund transfers
- fake identities in online verification
- manipulate visual evidence (e.g., altered CCTV footage)

Deepfake-based attacks are highly effective because they exploit trust in multimedia evidence (Rawat, 2025).



Figure 21: Deepfake based Attack

8.3.3 Psychology Behind Social Engineering

Key psychological principles used include:

- **Authority:** pretending to be a boss or official
- **Urgency:** your account will be locked in 1 hour
- **Scarcity:** fake offers or last-minute deals
- **Trust:** impersonating known individuals
- **Fear:** threats of account compromise or legal issues

Understanding these principles helps in analysing attacker motives and victims' responses (Rawat, 2025).

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8.4 Analysis of Simulated Attack

8.4.1 Identification of Suspicious Social Media Content

The first indicator of a potential attack was the Instagram story containing a video message claiming to announce a PayPal security upgrade. From an investigative perspective, several red flags were observed:

- The message created urgency, warning of possible account restriction
- It requested immediate verification, which is uncommon for official announcements
- It relied on a public figure endorsement rather than official PayPal communication channels

8.4.2 Analysis of the AI-Generated Video Content

To determine whether the video shared through the Instagram story was authentic or artificially generated, the video was analysed using multiple AI media detection tools. The objective of this stage was to identify synthetic media indicators and assess whether the video showed characteristics consistent with AI-generated content rather than a genuine recording.

- **Analysis Using Zhuque AI Detection Assistant**

The AI-generated video was submitted to the Zhuque AI Detection Assistant for forensic examination. This tool analyses visual patterns, facial consistency, and temporal artefacts commonly associated with synthetic media.

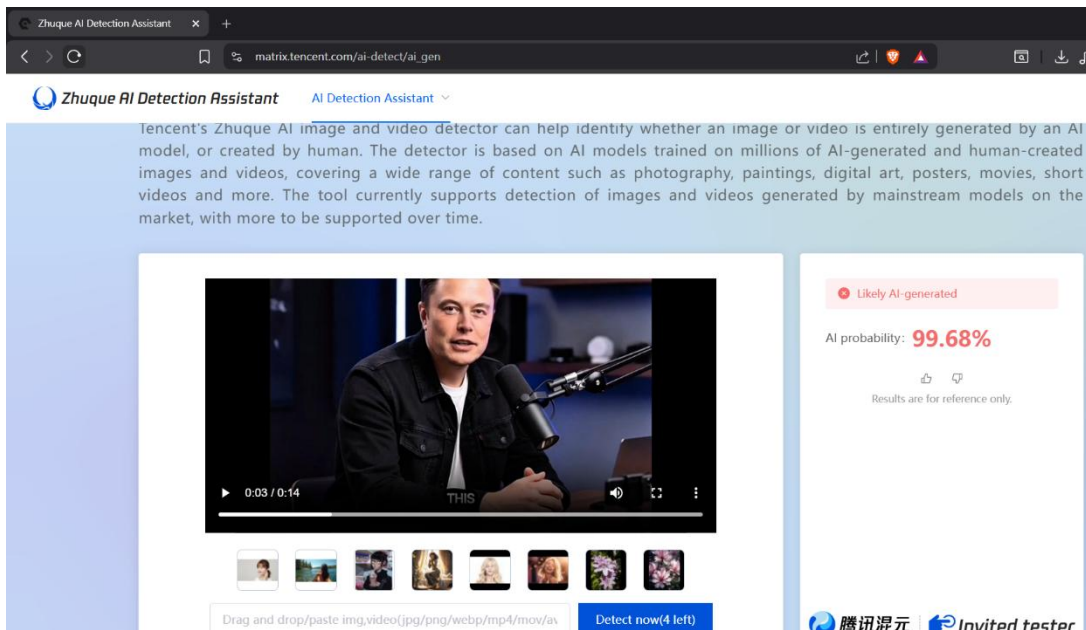


Figure 22: Results from 1st website

- **Analysis Using Sightengine**

To further validate the findings, the same video was analysed using Sightengine, an AI-based content moderation and synthetic media detection platform. Sightengine evaluates videos by analysing facial landmarks, motion coherence, and visual anomalies.

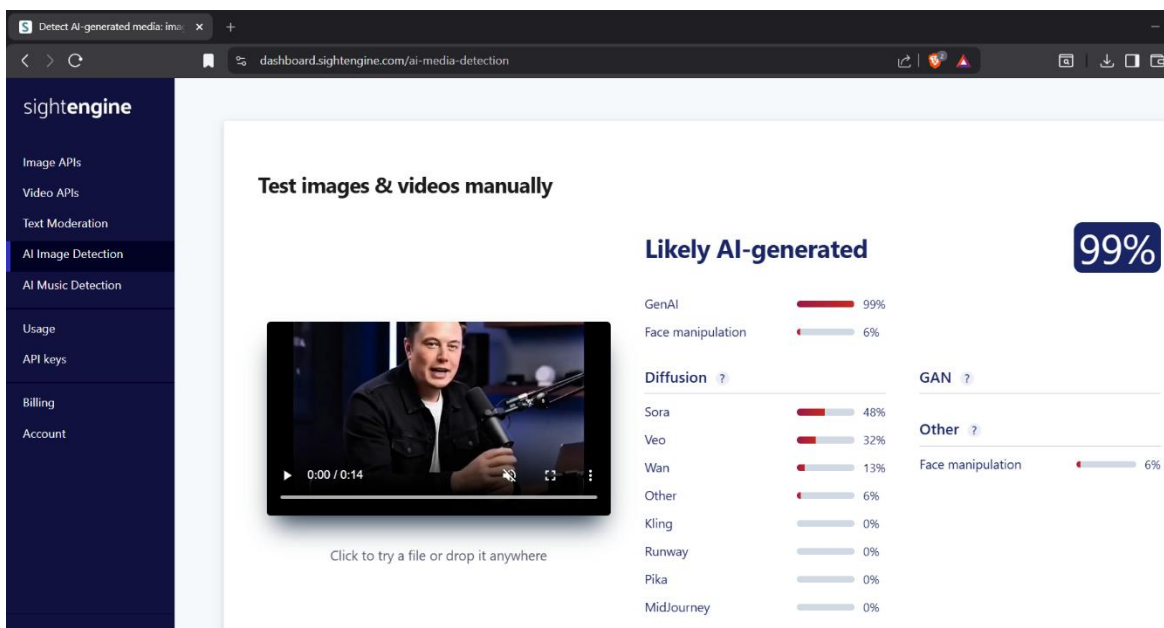


Figure 23: Results from 2nd website

8.4.3 Analysis of URL

At first glance, this URL “<https://paypal.com-verification@clck.ru/3RL98F>” appeared legitimate due to URL masking, which altered how the link was displayed to the user. This technique can conceal unfamiliar or suspicious domains, making users less likely to question the destination.

- **Analysis using urlscan.io**

Urlscan is widely used in digital investigations to verify whether a URL is associated with known phishing activity. The purpose of this analysis was to determine whether the URL had characteristics consistent with phishing websites and to compare it against verified legitimate domains.

The screenshot displays the urlscan.io analysis results for the URL <https://paypal.com-verification@clck.ru/3RL98F>. The main heading is **newer-summary-sim-anaheim.trycloudflare.com** with the IP address 104.16.230.132. A red banner indicates **Malicious Activity!** and a blue button says **Unlisted Scan**. The submitted URL is <https://clck.ru/3RL98F> and the effective URL is <https://newer-summary-sim-anaheim.trycloudflare.com/login.html>. The submission was on January 18 via manual (January 18th 2026, 5:02:29 pm UTC) from NP, scanned from DE. The page features a navigation bar with tabs: Summary, HTTP, Redirects, Links, Behaviour, Indicators, Similar, DOM, Content, API, and Verdicts. The **Summary** tab is active, showing that the website contacted 2 IPs in 2 countries across 3 domains to perform 2 HTTP transactions. The main IP is 104.16.230.132, located in Ascension Island and belongs to CLOUDFLARENET, US. The main domain is newer-summary-sim-anaheim.trycloudflare.com. The TLS certificate was issued by WE1 on December 14th 2025 and is valid for 3 months. It also notes that clck.ru was scanned 10000+ times on urlscan.io. The urlscan.io Verdict is **Potentially Malicious**. Targeting these brands includes PayPal (Financial). The **Live information** section shows the current DNS A record: 104.16.230.132 (AS13335 - CLOUDFLARENET, US). The **Screenshot** tab shows a PayPal login page. The **Page Title** is **Login - Paypal**. The **Page URL History** shows a redirect from <https://clck.ru/3RL98F> to <https://sbayandex.ru/redirect?url=https%3A%2F%2Fnewer-summary-sim-anaheim.trycloudflare.com&clien...>. The bottom navigation bar includes IP/ASNs, IP Detail, Domains, Domain Tree, Links, Certs, and Frames.

Figure 24: Fake URL result

The screenshot displays the urlscan.io analysis results for the website **www.paypal.com**. The browser address bar shows the URL `urlscan.io/result/019bd211-357f-75ec-b5d6-d8abf47a2aa3/`. The main content area is divided into several sections:

- Summary:** This section provides a high-level overview of the scan. It states that the website contacted 7 IPs in 2 countries across 5 domains to perform 87 HTTP transactions. The main IP is 151.101.193.21, located in the United States and belongs to FASTLY, US. The main domain is www.paypal.com. The Cisco Umbrella rank of the primary domain is 2745. The TLS certificate is issued by DigiCert EV RSA CA G2 on August 5th 2025, valid for a year. It also mentions that www.paypal.com has been scanned 10000+ times on urlscan.io.
- Verdict:** The urlscan.io Verdict is "No classification".
- Live information:** This section includes Google Safe Browsing status (No classification for www.paypal.com), current DNS A record (151.101.193.21), domain creation date (July 15th 1999), and domain registrar (MarkMonitor, Inc.).
- Domain & IP information:** This section provides details about the domain and IP address, including IP/ASNs, IP Detail, Domains, Domain Tree, Links, Certs, and Frames.
- Screenshot:** A screenshot of the website is shown, displaying the PayPal logo and the text "Bezahl, wie es dir passt".
- Page Title:** The page title is "PayPal Konto | Digitale Mobile Wallet | PayPal DE".
- Page URL History:** The page URL history shows the sequence of URLs visited during the scan, including `https://www.paypal.com/` and `https://www.paypal.com/de/home`.

Figure 25: Official URL result

8.4.4 Inspection Analysis of the Cloned PayPal Website

During HTML source-code inspection of the cloned PayPal login page, the following tag was identified:

```
<link rel="canonical" href="https://www.paypal.com/signin">
```

1. What Is a Canonical Link?

A canonical link is an HTML element used by search engines to indicate the preferred or "original" version of a webpage when duplicate or similar content exists. Legitimate websites use canonical tags to prevent SEO duplication issues.

2. Why Attackers Use Canonical Links in Phishing Pages

In the context of phishing and forensic investigation, the presence of a canonical link pointing to the real PayPal website is highly suspicious and reveals several attacker objectives:

- **Visual Authenticity**

By referencing the real PayPal URL, attackers attempt to increase the perceived legitimacy of the cloned page, especially if the source code is inspected casually.

- **Search Engine Evasion**

Search engines may associate the phishing page with the legitimate PayPal domain, reducing the likelihood of immediate blacklisting.

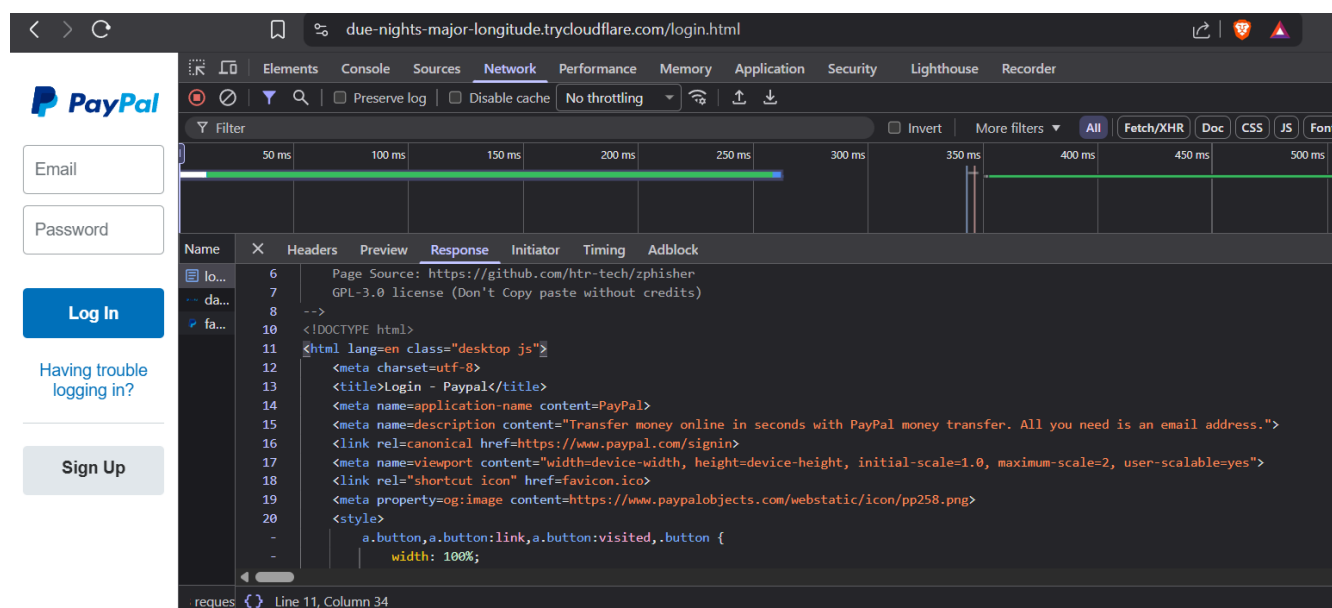


Figure 26: SS of tag

8.4.5 IP Address Analysis of Cloned vs Legitimate PayPal Website

To further validate the legitimacy of the cloned PayPal website, DNS and IP address analysis was performed using nslookup.io. DNS analysis is a fundamental digital forensic technique used to determine domain ownership, hosting infrastructure, and trust relationships. By comparing the network-level properties of the cloned site with the real PayPal website, investigators can identify impersonation, infrastructure abuse, and malicious hosting patterns.

- **Analysis of the Cloned PayPal Website**

Using nslookup.io, the domain resolved to IP addresses associated with Cloudflare's tunnelling and reverse proxy infrastructure rather than a

dedicated PayPal-owned network. This indicates that the cloned PayPal page was hosted via a Cloudflare Tunnel, a service commonly abused by attackers to quickly deploy phishing sites without registering suspicious domains.

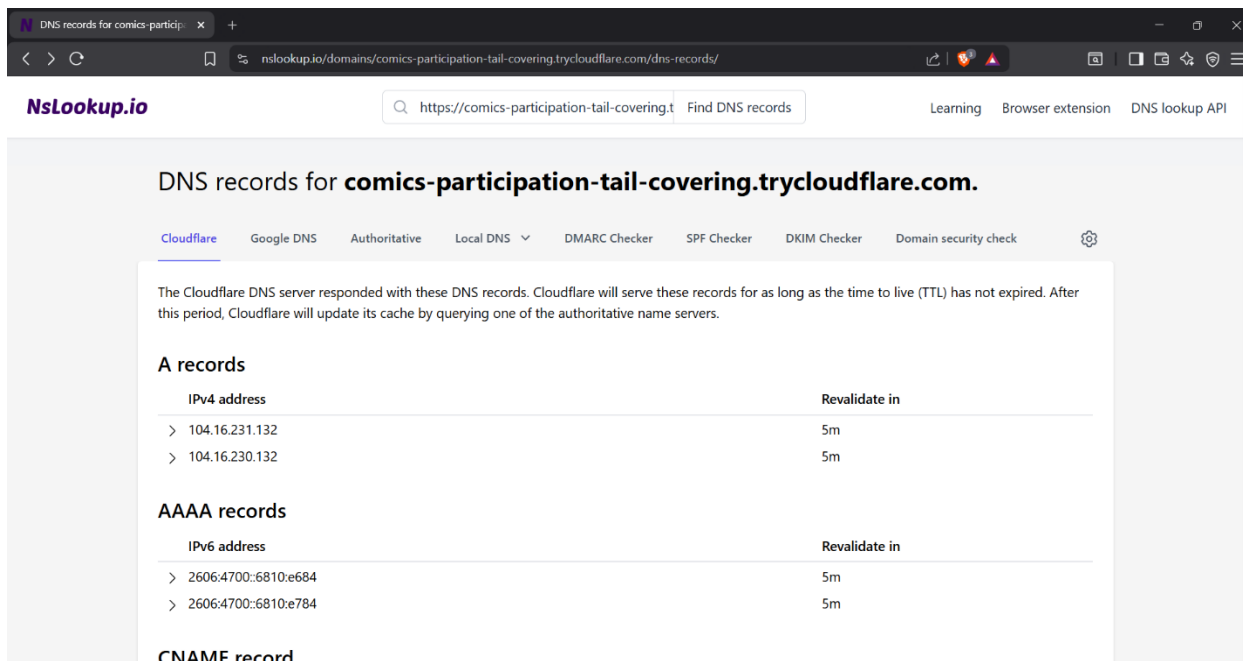


Figure 27: Analysis of Cloned Webpage

- **Analysis of the Legitimate PayPal Website**

DNS resolution of the legitimate PayPal website revealed IP addresses belonging to PayPal's official, globally distributed infrastructure, protected by enterprise-grade CDNs and security controls. This confirms that the legitimate PayPal website operates on dedicated, trusted infrastructure, unlike the cloned phishing site.

DNS records for **www.paypal.com**.

The Cloudflare DNS server responded with these DNS records. Cloudflare will serve these records for as long as the time to live (TTL) has not expired. After this period, Cloudflare will update its cache by querying one of the authoritative name servers.

A records

IPv4 address	Revalidate in
> 151.101.129.21 For CNAME paypal-dynamic.map.fastly.net--www.glb.paypal.com.	3m 24s
> 151.101.1.21 For CNAME paypal-dynamic.map.fastly.net--www.glb.paypal.com.	3m 24s
> 151.101.193.21 For CNAME paypal-dynamic.map.fastly.net--www.glb.paypal.com.	3m 24s
> 151.101.65.21 For CNAME paypal-dynamic.map.fastly.net--www.glb.paypal.com.	3m 24s

AAAA records
No AAAA records found.

CNAME record

Figure 28: Analysis of Legitimate Website

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